

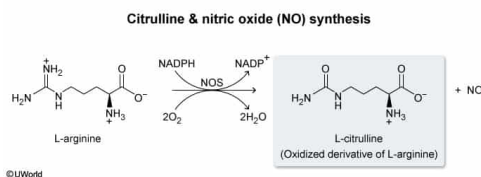
Citrulline is an important component in the urea cycle and in the cellular production of nitric oxide (NO). When L-arginine is oxidized via a nitric oxide synthase (NOS) enzyme, it results in the formation of L-citrulline and the release of NO(g). Which structure most likely corresponds to citrulline?

- ☐ A. [5%]
- ☐ B. [15%]
- ☐ C. [17%]
- ☒ D. [61%]

Correct

60% answered correctly

Explanation:



Amino acids are organic molecules that play an important role in many biological processes. The **structure** of amino acids has three components bonded to the same **carbon** (known as the **α -carbon**): a **carboxyl** group, an **amino** group, and a **side chain**. There are 20 **common amino acids** that differ only in the side chain, which contains different functional groups. The amino acid side chains often undergo chemical reactions to produce derivatives of the respective amino acid.

Arginine has a side chain with three aliphatic carbon atoms and a three-nitrogen guanidine group. The nitric oxide synthases (NOS) oxidize arginine at the **guanidinium group**, converting an amine to an **amide** and releasing NO(g) in the process. This oxidized derivative of arginine is called citrulline. Citrulline has the same number of carbons and the same stereochemistry as arginine.

(Choice A) This is the structure of **serine**. It does not have the same number of carbon atoms as arginine, and an alcohol is not the oxidation product of arginine.

(Choice B) This is the structure of **glutamine**, which does not have the same number of carbon atoms as arginine.

(Choice C) This is the structure of **lysine**, which has three aliphatic carbon atoms like arginine, but a **primary amine** is not the oxidation product of arginine.

Educational objective:

Amino acids play an important role in biological processes. They have three substituents bonded to the same carbon (α -carbon): a carboxyl group, an amino group, and a side chain. Amino acids are distinguished by their side chains, which have functional groups that can participate in a reaction.

Time spent:0 sec**Subject:**
Organic Chemistry**QID:**402775**System:**
Functional Groups and Their Reactions